

Turning trace on

Waves require two cooperating pieces

Flag plus dump calls

- Enable trace at compile with the trace or trace-fst flags as your flow requires
- In the host, create a trace pointer, open a file, call dump each eval step you care about,
- Dump without the compile flag is BLIND, you wrote files the model never instrumented
- Both sides must agree

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Verilator trace

Document-aid literacy: `--trace` / `--trace-fst` and C++ TB dump calls → VCD/FST roles (not a real Verilator run). Starter: `--trace · VCD · open+dump — READY`.

Starter example: `--trace · VCD · open+dump+close — READY`.

Load starter example

Challenges 0 / 22 cleared

Quiz: `--trace`: `--trace` is...

☐ a compile-time flag that enables tracing support in the Verilated model

☐ a GTKWave cursor name

☐ the same as +SEED

☐ only a Makefile PHONY target

Show hint

Check

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Idle

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Core ideas

--trace
Compile-time enable for wave models.

--trace-fst
Same path, FST instead of VCD.

tftp->dump
TB must sample time into the file.

VCD / FST
Portable text vs compact binary dump.

Lab

SCENARIO

Lab starter

Real Verilator practice

- In Track A, run a tiny design with trace enabled, produce a wave file
- Point to the host lines that open and dump
- Delete dump temporarily and notice OPEN behavior; compare to a correct run
- Keep the file small, this module is trace literacy, not a farm dump

Pitfalls to watch

- Do not enable trace and forget dump in the eval loop
- Do not dump when compile never enabled trace, BLIND files mislead you
- Do not leave trace files unclosed on long runs, flush and close matter
- And do not open gigabyte waves when a short window suffices

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, fix OPEN and BLIND scenarios to ready
- In Track A, generate one viewable trace from a short run
- When you are ready, take the short quiz, then continue to wave dump literacy